IGO GROUP ERP

Order-to-Delivery Module Plan

A complete redesign story — from Sales Order through Purchase, Hub Receiving, Delivery & Finance.

Project	**FFERPv2 — Centralized ERP Redesign**
Prepared for	**IGO Group · FF GM · CEO**
Modules	**10 Core Modules**
Stack	**React · Supabase · React Native · WhatsApp API**
Date	**22 May 2026**
Status	**Architecture Phase — Pre-Build**

About This Document

This document defines the architecture and feature plan for the Order-to-Delivery system redesign within the IGO Group ERP (FFERPv2). It covers 10 modules spanning the complete daily business cycle — from the moment a customer places an order on any channel, through procurement, hub receiving, delivery, and final vendor payment approval.

The design is driven by a single operational story: the BD team collects orders from 10 AM to 11 PM across website, mobile app, WhatsApp, and manual entry. At 11:59 PM every night, the system automatically generates Purchase Orders per hub based on what was ordered vs. what is in stock. By 12:00 AM the Purchase Executive has their PO ready. Boxes arrive at the hub with barcodes. The Hub Manager scans them in. Delivery goes out. The vendor bill flows through FF GM → Admin → CEO for approval. The cycle repeats.

ORDER WINDOW	HUBS	PO TRIGGER	CHANNELS
10:00 AM – 11:00 PM	**3 Hubs Geo -Assigned**	**11:59 PM Automatic**	**4 Sources Centralized**

System Flow Overview

①	**Order Collection**	Website · App · WhatsApp · Manual → Central Queue
②	**Hub Assignment**	Geo-distance engine assigns nearest hub + manual override
③	**GM Dashboard**	Hourly order feed · Team targets · PO preview · Approval
④	**EOD PO Engine**	11:59 PM auto-fire · Inventory delta calc · Hub-wise POs
⑤	**Purchase Executive**	Shift-based PO routing · Vendor procurement · Box labeling
⑥	**Barcode System**	QR/Barcode per box · Item · Qty · Hub · PO Reference
⑦	**Hub Scanner App**	React Native · Scan arrival · QC · Wastage · Realtime sync
⑧	**Inventory Engine**	Live stock per hub · Auto +/– on scan · PO Engine feeds from here
⑨	**Delivery Module**	Pack barcodes · Dispatch scan · Driver confirm · Customer closed
⑩	**Bill Approval**	Auto bill → FF GM → Admin → CEO → Vendor payment

MODULE 01 FOUNDATION LAYER

Centralized Order Hub

"All channels. One queue. Zero duplication."

Unified order intake from all 4 channels — Website, Mobile App, WhatsApp (Meta Business API + manual fallback), and direct ERP form entry. Every order lands in a single normalized queue with full source attribution.

KEY FEATURES

- → Website orders via REST webhook — auto-synced on checkout
- → Mobile app orders via Supabase Realtime — zero latency
- → WhatsApp Business API parses free-text orders automatically
- → Manual fallback panel for BD team to enter WhatsApp/phone orders
- → Unified order schema: Source Tag · Customer ID · Creator ID · Timestamp · Status
- → Duplicate detection engine — prevents double-entry across channels

TECHNOLOGY

Supabase Realtime | **WhatsApp Business API** | **Webhook Gateway** | **Edge Functions**

MODULE 02 ORDER PROCESSING

Geo Hub Assignment Engine

"Right hub. Right zone. Every time."

Automatic hub assignment based on customer delivery address. The engine geocodes the address, calculates distance to each of the 3 hubs, and assigns the nearest one. BD team can manually override the assignment when required.

KEY FEATURES

- → Address geocoding via Maps API — converts address to coordinates
- → Distance calculation to all 3 hubs — nearest wins
- → 3-hub zone mapping with configurable radius boundaries
- → Manual override toggle for BD team — with reason logging
- → Hub assignment locked on order confirmation — no silent changes
- → Hub re-assignment audit trail for every change

TECHNOLOGY

Geocoding API | **Supabase PostGIS** | **Hub Zone Config** | **Audit Log**

MODULE 03

MANAGEMENT LAYER

FF GM Command Dashboard

"Full visibility. One screen. Hourly pulse."

The nerve center for the FF General Manager. Live order feed refreshed hourly, per-employee sales target tracking, hub-wise order breakdown, and a one-click PO approval button. The GM controls when the day ends or lets automation handle it.

KEY FEATURES

- → Per-employee target card: daily target vs. achieved vs. remaining
- → Live order feed — refreshes every hour, shows creator name per order
- → Hub-wise order summary: Hub 1 / Hub 2 / Hub 3 totals side by side
- → PO Preview panel — shows pending PO quantities before EOD
- → Early Approval button — triggers immediate PO generation on click
- → If no approval: system auto-fires PO at 11:59 PM without action

TECHNOLOGY

TanStack Query | **Supabase Realtime** | **Edge Function Trigger** | **Role-gated UI**

MODULE 04

AUTOMATION CORE

EOD Purchase Order Engine

"11:59 PM. Every night. No exceptions."

A scheduled Edge Function fires at 11:59 PM and generates one Purchase Order per hub. It reads all confirmed orders for the day, cross-references live inventory, calculates the exact shortfall, and locks the PO. If the GM approved early, it fires immediately at approval time instead.

KEY FEATURES

- → Cron-scheduled Edge Function — fires at 23:59 daily, no manual trigger needed
- → PO quantity = Total Order Qty – Current Hub Stock (per item)
- → Separate PO generated per hub — Hub 1, Hub 2, Hub 3 independently
- → GM early-approval triggers instant PO — bypasses the 11:59 timer
- → PO locked after generation — no edits, only amendment requests
- → PO pushed immediately to Purchase Executive queue at 00:00

TECHNOLOGY

Supabase Cron | **Inventory API** | **Edge Functions** | **PO Lock System**

MODULE 05

PROCUREMENT LAYER

Purchase Executive Portal

"Right PO. Right person. Right hub."

At 12:00 AM, the system automatically routes each hub's PO to whichever shift user is logged into that hub. The Purchase Executive sees their assigned PO, goes to the vendor, procures items, and triggers barcode generation per box.

KEY FEATURES

- → Shift-aware routing — PO assigned to the shift user logged into Hub X
- → PO arrives in executive's day-start queue at 00:00 exactly
- → Full PO detail view: item list, quantities, vendor info, hub destination
- → In-app vendor confirmation — executive marks each item as procured
- → Procurement completion triggers automated barcode/QR generation per box
- → Late or partial procurement flagged — GM and Admin notified

TECHNOLOGY

Shift Auth System | **Supabase Realtime** | **PO Router** | **Push Notifications**

MODULE 06 TRACEABILITY LAYER

Barcode / QR Generation System

"Every box. Fully traceable. Scan to know."

On PO fulfillment, the system generates a unique QR code or barcode for every physical box. Each code encodes item identity, quantity, hub destination, and PO reference — making every box fully traceable from vendor to customer.

KEY FEATURES

- → Auto-generated per box on procurement completion — no manual step
- → Encodes: Item Name · Quantity · Hub Destination · PO Reference · Date · Vendor ID
- → QR Code (camera-scannable) or Barcode (hardware-scannable) — configurable
- → Printable label generated as PDF — ready for physical printing on box
- → Each code links back to live ERP data — scan reveals full chain history
- → Bulk generation for large POs — all boxes labeled in one action

TECHNOLOGY

QR/Barcode Library | PDF Label Generator | ERP Data Link | Print API

MODULE 07

HUB OPERATIONS LAYER

Hub Manager Scanner App

"Scan. Receive. Update. Done."

A React Native mobile app for iOS and Android used exclusively by Hub Managers. At unloading, the manager scans each box barcode — the app reads the item and quantity, updates inventory in real time, and guides QC and wastage logging.

KEY FEATURES

- → React Native (iOS + Android) — single codebase, both platforms
- → Camera-based barcode / QR scan — no extra hardware required
- → Box arrival scan → ERP inventory auto-incremented instantly
- → QC pass/fail per item — with photo evidence capture in-app
- → Wastage logging — reason codes, quantity, photo attached
- → All actions sync to ERP in real time — hub and office see same data

TECHNOLOGY

React Native | **Expo Camera** | **Supabase Realtime** | **Offline Queue**

MODULE 08

INVENTORY LAYER

Smart Inventory Management

"Stock is live. Always accurate. Hub by hub."

Inventory is the connective tissue of the entire system. Stock levels update automatically on box arrival, delivery dispatch, wastage, and QC rejection. The EOD PO Engine reads from inventory to calculate exactly what needs ordering.

KEY FEATURES

→ Per-hub stock ledger — Hub 1, Hub 2, Hub 3 tracked independently

→ Stock increases on: box arrival scan (+)

→ Stock decreases on: delivery dispatch scan (–) and wastage log (–)

→ QC rejections auto-deducted with rejection reason linked

→ PO Engine reads live stock before generating purchase quantities

→ Low-stock alerts — notifies Hub Manager and GM when threshold breached

TECHNOLOGY

Supabase DB | **Stock Ledger API** | **Alert Engine** | **Inventory Dashboard**

MODULE 09

FULFILLMENT LAYER

Delivery & Distribution Module

"Pack. Scan. Dispatch. Delivered."

Each customer delivery pack receives its own barcode linked to the order ID. Scanning on dispatch decrements inventory and marks the order as out for delivery. Driver or hub confirmation closes the order loop completely.

KEY FEATURES

- → Delivery pack barcode auto-generated and linked to customer order ID
- → Dispatch scan → inventory decremented + order status → 'Out for Delivery'
- → Route-wise pack grouping — all packs for one route batched together
- → Driver app or hub app confirms delivery — customer notified automatically
- → Partial delivery handling — remaining items tracked for next dispatch
- → Daily delivery summary report — packs dispatched, delivered, pending

TECHNOLOGY

Barcode Engine | **Driver App** | **Inventory API** | **Customer Notifications**

MODULE 10

FINANCE LAYER

PO Bill & Approval Workflow

"Auto-generated. Three-step approved. Vendor paid."

After delivery completion the system auto-generates the vendor bill from the PO. It enters a 3-step approval chain: FF GM reviews, Admin validates, CEO gives final approval. Once cleared, vendor payment is triggered. This cycle repeats every day.

KEY FEATURES

- → Vendor bill auto-generated from fulfilled PO data — zero manual entry
- → Step 1 — FF GM: reviews quantities and amounts, approves or flags
- → Step 2 — Admin: validates bill against procurement records, signs off
- → Step 3 — CEO: final approval with full bill summary visible
- → Each approver notified via push + in-app alert — no chasing needed
- → On CEO approval: payment order issued and vendor ledger updated

TECHNOLOGY

Bill Generator | **Approval Chain Engine** | **Push Notifications** | **Vendor Ledger**

Next Steps & Open Questions

Before build begins, the following items need confirmation to finalise the database schema and API design for each module.

1	**Hub Names & Zones** What are the actual names of Hub 1, 2, and 3? Do you have pincode ranges or locality lists for each zone boundary already?
2	**WhatsApp Order Format** When a customer sends a WhatsApp order, what does it look like? Free text ("2kg tomatoes, 1kg onions") or a structured catalogue reply? This determines how the parser is built.
3	**Inventory Starting Point** Is inventory currently tracked in the ERP already, or are we building the stock ledger from scratch with this module?
4	**Barcode Format** QR Code (camera-scannable, any phone) or traditional Barcode (hardware scanner)? Or both? This affects the hub app scanner implementation.
5	**Vendor Bill Delivery** Is the PO bill a PDF sent via email to the vendor, or does it also appear in the vendor portal inside the ERP? Or both?
6	**Build Priority** Which module do you want to start with first? Recommended: Module 01 (Centralized Order Hub) as the foundation for all downstream modules.

■ This entire order-to-delivery cycle is designed to run automatically every single day. Once all 10 modules are live, the system requires zero manual triggers for the core workflow.